

Executive Summary

Corrected Technical Position

The notebook now uses leakage-safe cumulative historical CTR features. Training rows only see previous impressions; validation rows use development history; test rows use all training history.

Final Model

Logistic Regression - balanced selected by highest tuned-threshold F1. Threshold 0.488; ROC-AUC 0.598; PR-AUC 0.082; F1 0.144; recall 0.461; precision 0.085.

Performance Interpretation

The model provides moderate discrimination, not high accuracy. It should be treated as a ranking/targeting aid rather than a standalone automated decision system.

Business Use

Use predicted probabilities, product CTR, segment CTR, and weekend uplift for bid testing, creative prioritization, and inventory planning.

Methodology Corrections

- ? Removed target leakage from aggregate CTR features by replacing full-development target maps with cumulative past-only training features.
- ? Added always-non-click and global-CTR baselines to show the starting point for an imbalanced classifier.
- ? Added feature ablation: raw+temporal baseline, interaction history counts, and leakage-safe historical CTR features.
- ? Corrected SMOTE language: no one-hot interpolation is used; SMOTENC should be used if categorical-aware oversampling is added later.
- ? Added two-proportion weekend test and confidence intervals for high-propensity user profiles.
- ? Replaced broad feature grouping with top individual feature names.

Model Comparison ? Full Temporal Holdout

	model	threshold	roc_auc	pr_auc	f1	precision	recall	tn	fp	fn	tp
Logistic Regression	- balanced - tune	0.4881	0.5979	0.0818	0.1436	0.0851	0.4606	45268	21797	2374	2027
Random Forest	- balanced - tune	0.4881	0.5977	0.0821	0.1432	0.0839	0.4897	43531	23534	2246	2155
Random Forest	- balanced - tune	0.5000	0.5977	0.0821	0.1429	0.0857	0.4281	46975	20090	2517	1884
Logistic Regression	- balanced - tune	0.5000	0.5979	0.0818	0.1413	0.0862	0.3915	48806	18259	2678	1723
XGBoost	- weighted - tune	0.5000	0.5936	0.0829	0.1405	0.0817	0.5019	42221	24844	2192	2209
XGBoost	- weighted - tune	0.5000	0.5936	0.0829	0.1405	0.0817	0.5019	42221	24844	2192	2209
Global CTR	probability	0.0687	0.5000	0.0616	0.1160	0.0616	1.00	0	67065	0	4401
Always non-click	baseline	0.5000	0.5000	0.0616	0.0000	0.0000	0.0000	67065	0	4401	0

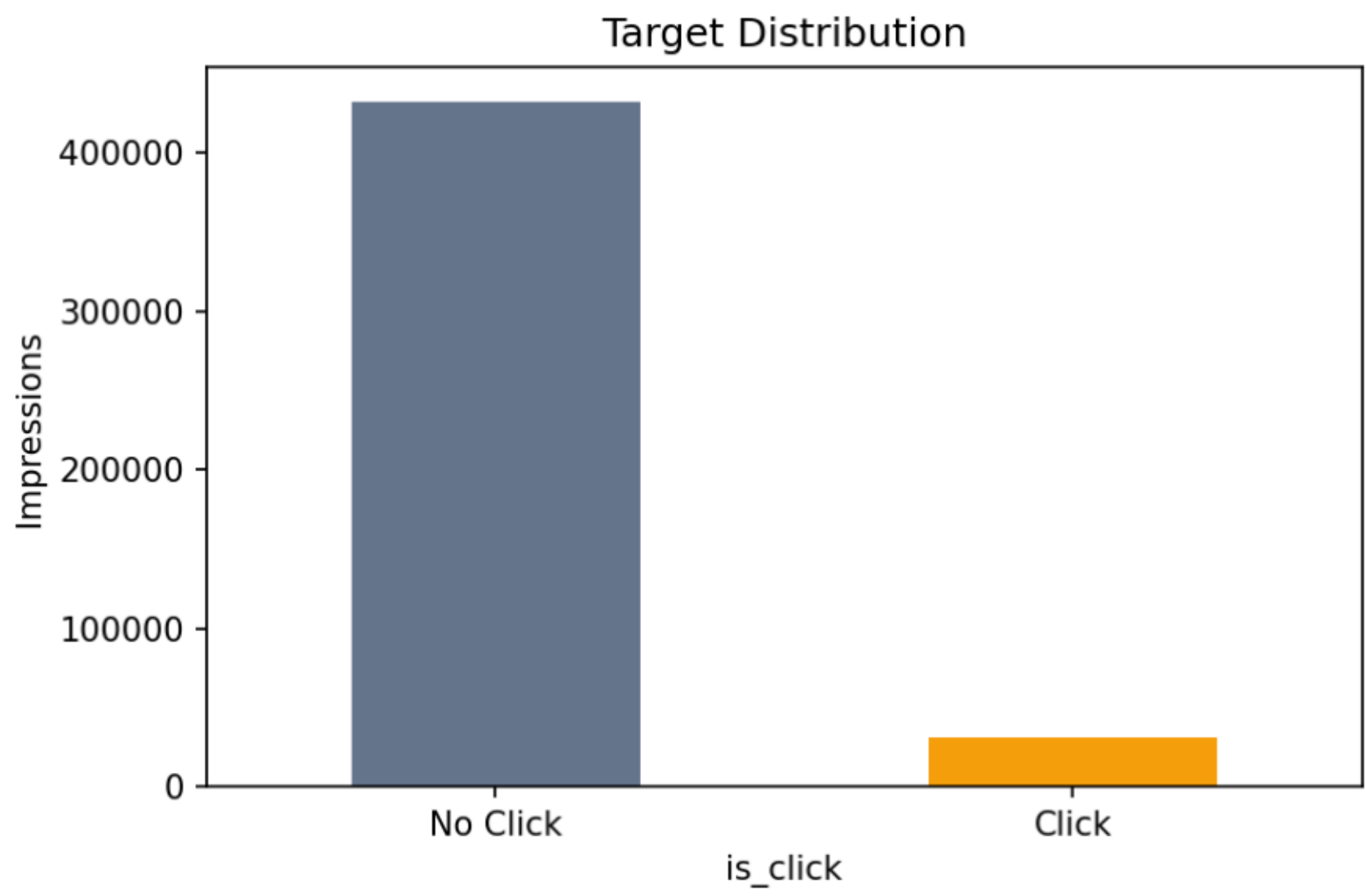
Selection is based on F1 balance, while ROC-AUC and PR-AUC leaders are still disclosed.

Feature Ablation ? Incremental Value

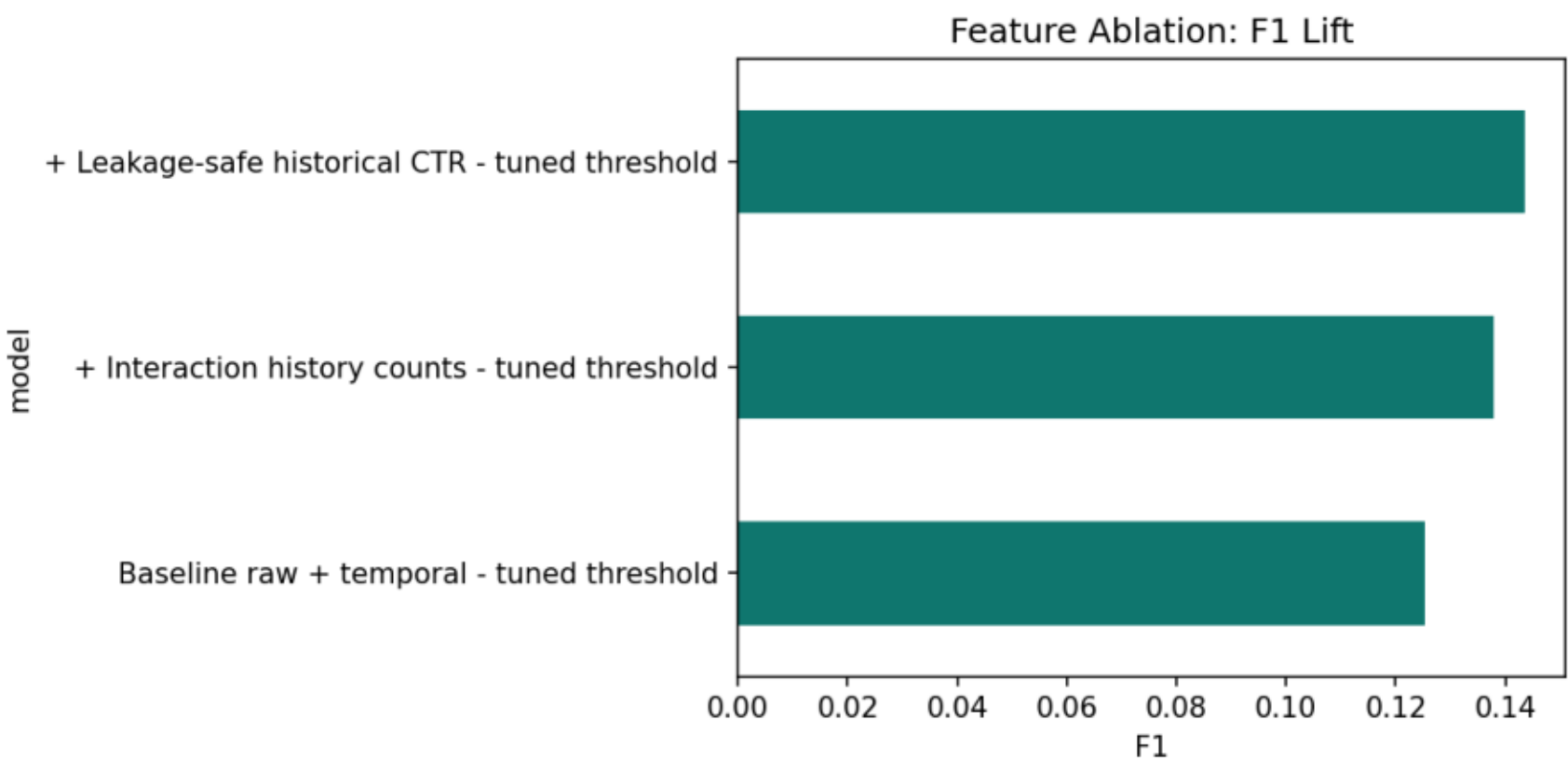
	model	threshold	roc_auc	pr_auc	f1	precision	recall
Baseline raw + temporal - tuned		0.4762	0.5491	0.0703	0.1253	0.0707	0.5476
+ Interaction history counts - tune		0.4762	0.5902	0.0792	0.1379	0.0792	0.5319
+ Leakage-safe historical CTR - tune		0.4881	0.5979	0.0818	0.1436	0.0851	0.4606

This directly tests whether interaction and historical CTR features add predictive value.

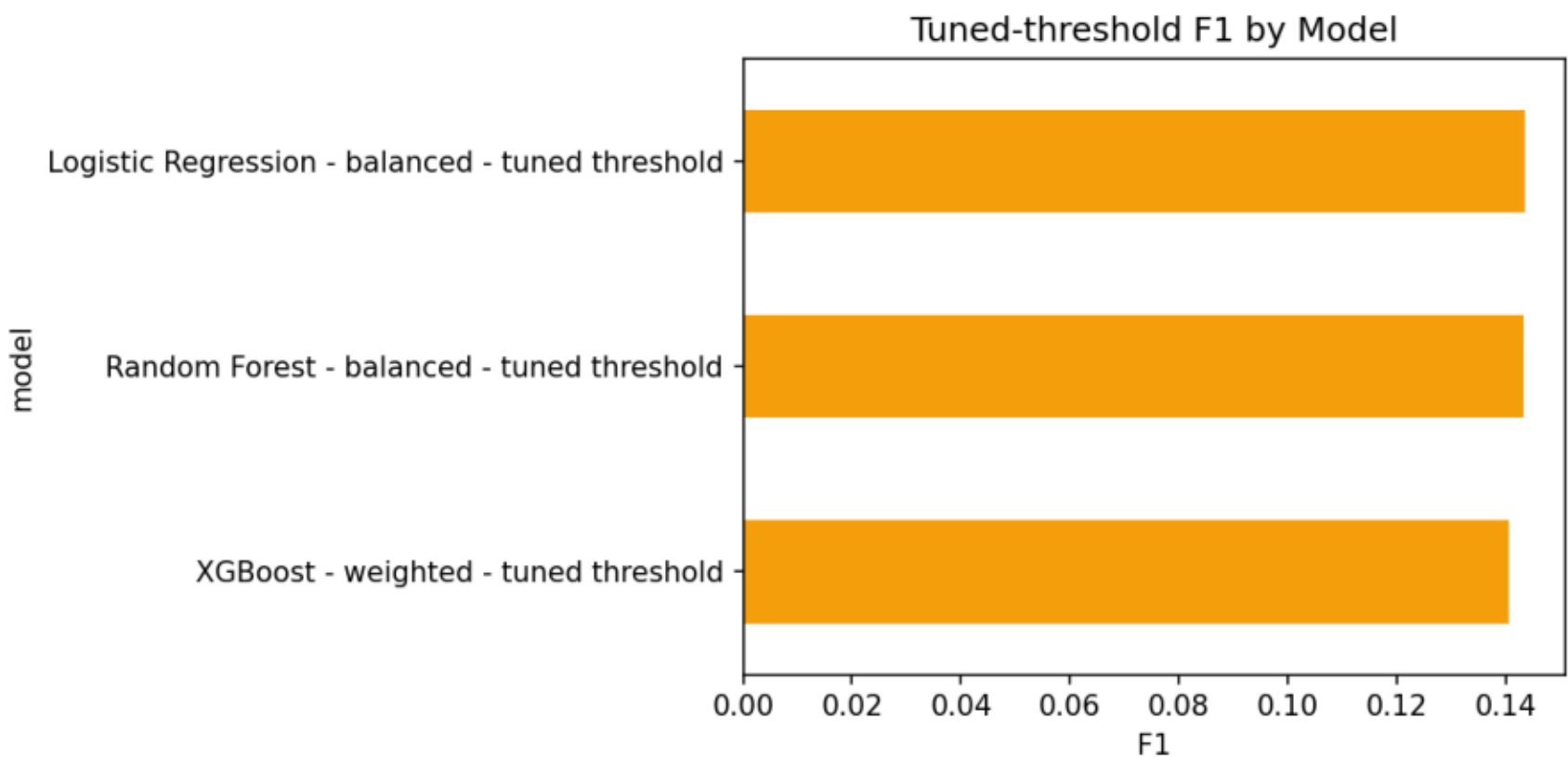
Target Distribution



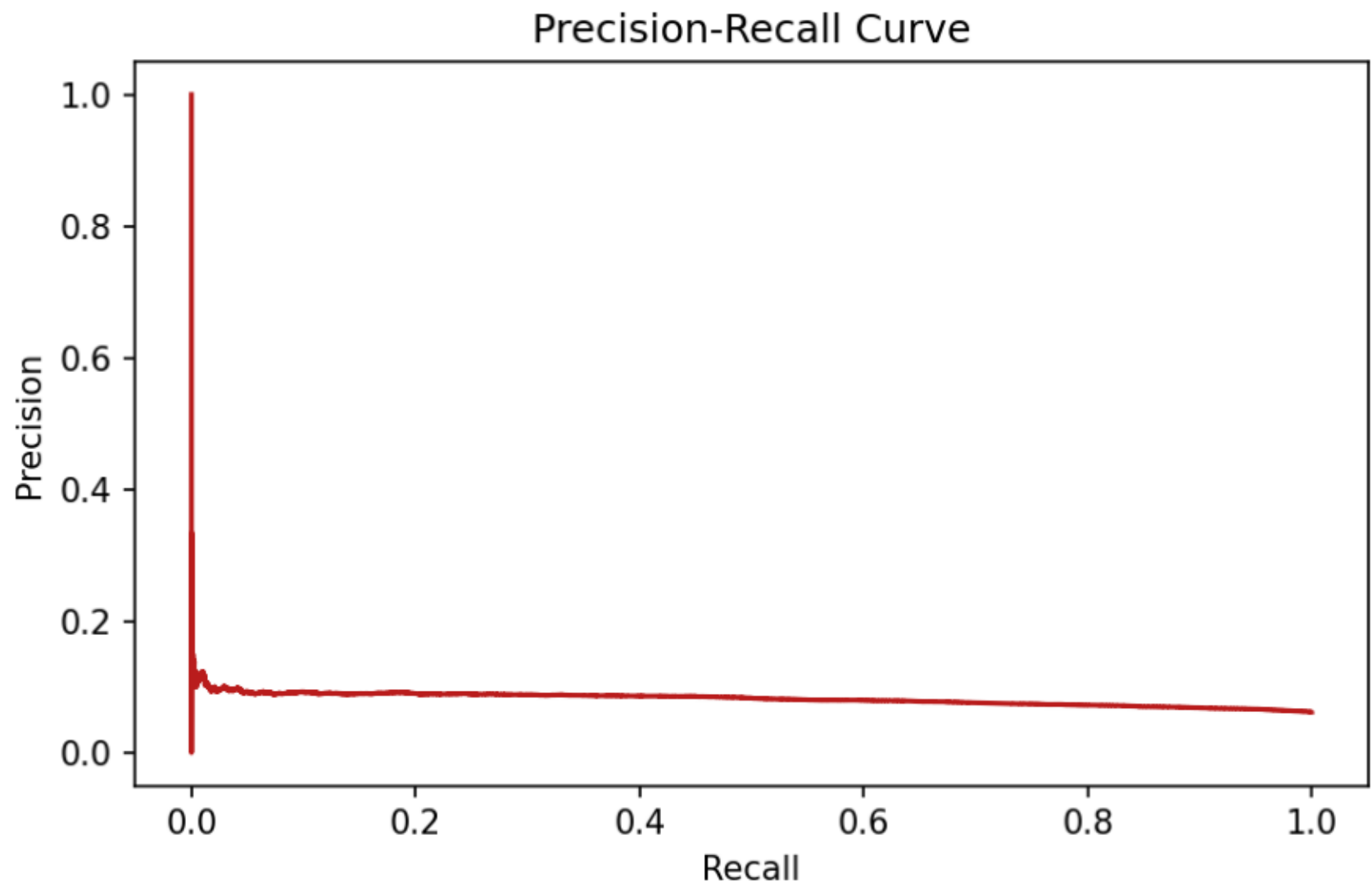
Feature Ablation Chart



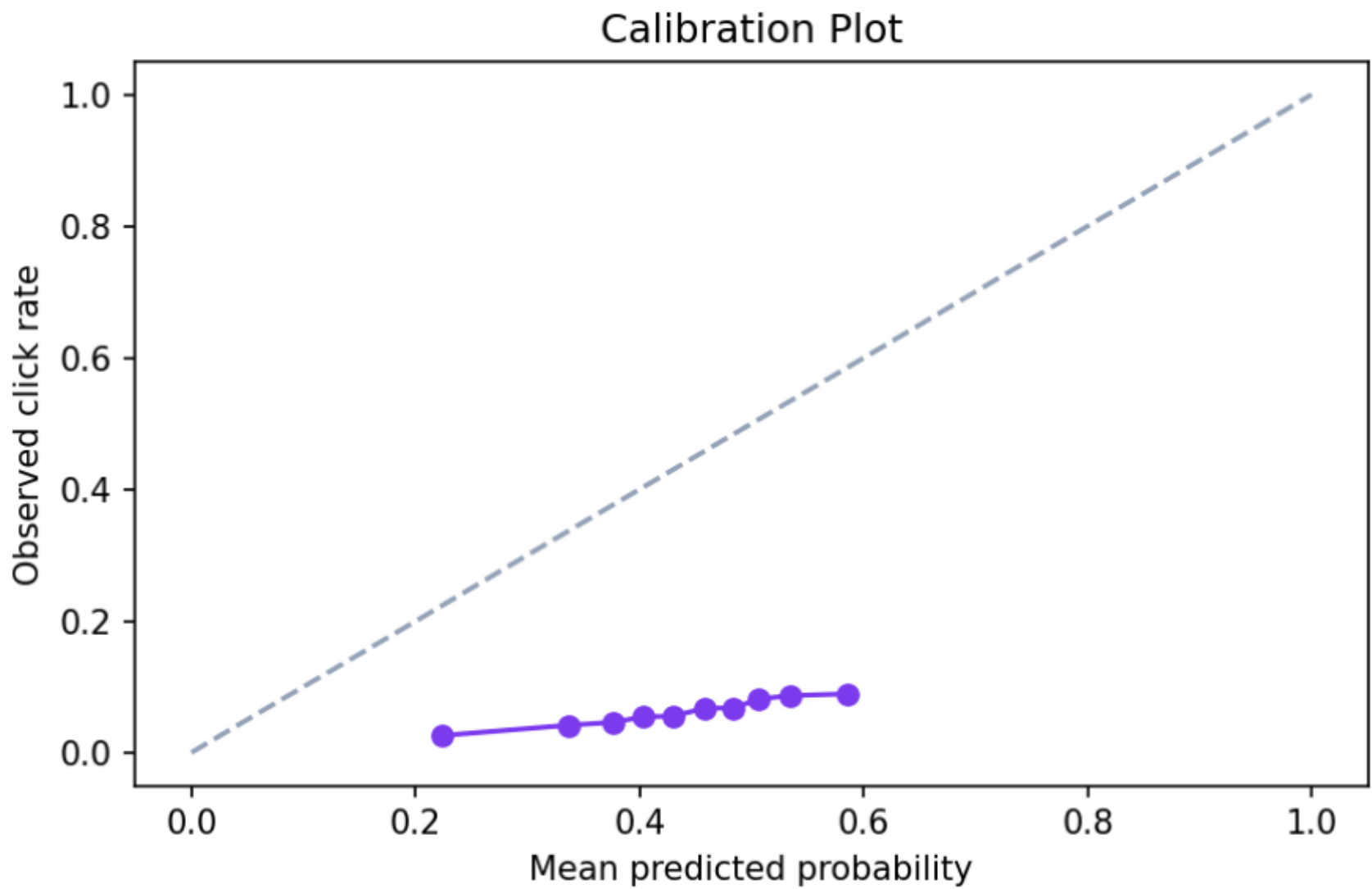
Model F1 Comparison



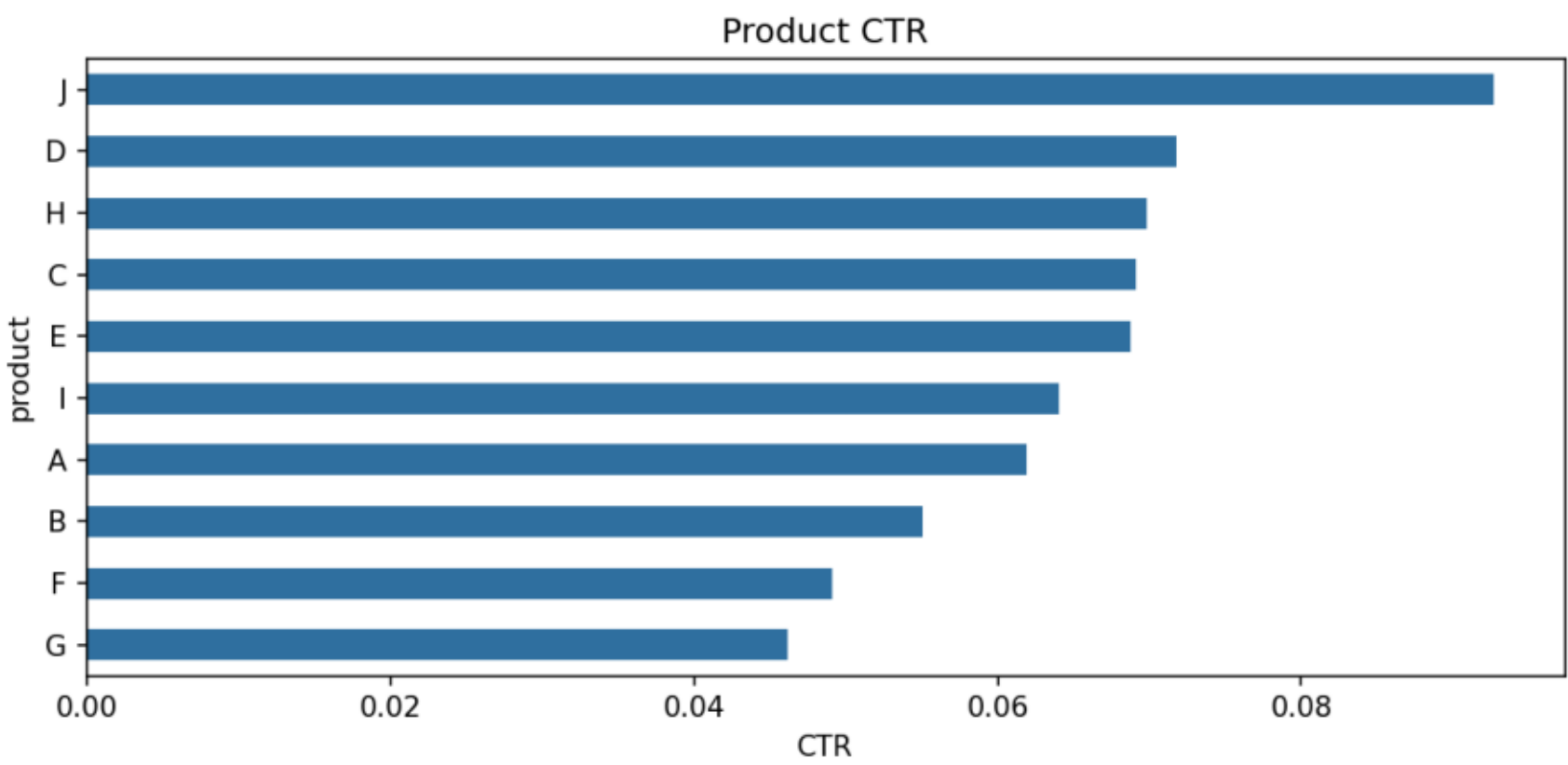
Precision-Recall Curve



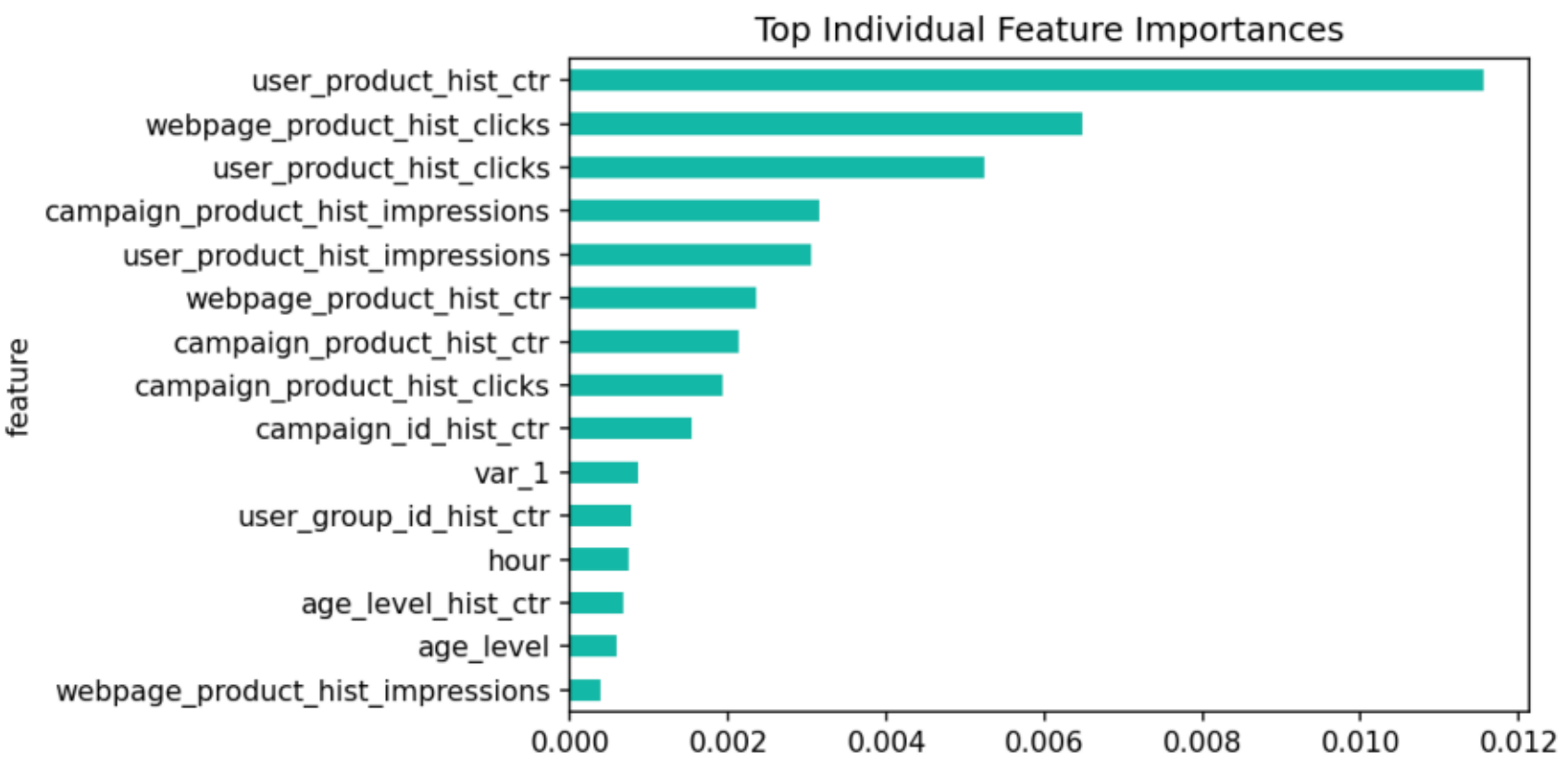
Calibration Plot



Product CTR / Inventory Planning Signal



Top Individual Feature Importances



Top Product CTR Planning Signal

product	clicks	impressions	ctr	expected_clicks_per_100k_impressions
J	899	9698	0.0927	9,269.95
D	2949	41064	0.0718	7,181.47
H	7654	109574	0.0699	6,985.23
C	11306	163501	0.0691	6,914.94
E	1474	21452	0.0687	6,871.15
I	4079	63711	0.0640	6,402.35
A	953	15391	0.0619	6,191.93
B	1238	22479	0.0551	5,507.36
F	344	7007	0.0491	4,909.38
G	435	9414	0.0462	4,620.78

This is an inventory-planning signal, not a formal demand forecast.

High-Propensity User Profiles with Wilson Intervals

gender	age_level	city_development_index	clicks	impressions	ctr	ctr_ci_low	ctr_ci_high
Male	5.00	4.00	255	2945	0.0866	0.0770	0.0973
Female	5.00	4.00	100	1181	0.0847	0.0701	0.1019
Male	1.00	2.00	605	7922	0.0764	0.0707	0.0824
Female	5.00	2.00	208	2752	0.0756	0.0663	0.0861
Male	2.00	2.00	3736	50409	0.0741	0.0719	0.0764
Male	1.00	3.00	607	8291	0.0732	0.0678	0.0790
Male	1.00	4.00	306	4252	0.0720	0.0646	0.0801
Male	5.00	3.00	349	4909	0.0711	0.0642	0.0786
Male	2.00	3.00	1927	27671	0.0696	0.0667	0.0727
Male	5.00	2.00	493	7112	0.0693	0.0636	0.0755

ML Insight to Business Action

Model insight	Business action
High product CTR	Increase exposure and reserve more inventory
High webpage CTR	Prioritize placement and monitor page quality
High user-product historical CTR	Personalize ad selection and bid higher
Weekend uplift	Test weekend bid modifier before permanent adoption
High predicted probability	Prioritize impression under budget constraints
Low CTR product	Refresh creative or reduce allocation
Model drift	Retrain and recalibrate thresholds

Operational recommendations map model evidence to ad-serving decisions.

Business Questions ? Corrected Answers

? Weekend users: weekend CTR is 7.33% vs weekday CTR 6.65%; uplift is 0.68 percentage points / 10.2% relative; p-value $4.27e-12$.

? Product performance: highest CTR products are J, D, H; lowest CTR products are B, F, G.

? Personalization: ablation shows interaction history counts and leakage-safe CTR aggregates improve PR-AUC/F1 over the raw+temporal baseline.

? Drivers: top individual features and CTR history signals should inform placement prioritization, personalization, and creative allocation.

? SMOTE: not used in final training because categorical-aware SMOTENC is required; class weighting is simpler and avoids synthetic categorical records.

? Inventory: historical product CTR becomes expected clicks per 100K impressions, a planning signal for ad inventory allocation.

? Profiles: bid adjustments should use minimum-volume filters, confidence intervals, fairness/privacy checks, and drift monitoring.

Limitations and Next Steps

- ? Current performance is below the approach-document aspirational targets for ROC-AUC/F1/precision.
- ? Add richer behavioral history, creative metadata, bid price, recency, placement quality, and conversion value features if available.
- ? Calibrate probabilities before automated budget allocation.
- ? Run live A/B tests before converting offline CTR gains into permanent bidding rules.
- ? Use SMOTENC or class-weighted gradient boosting experiments if categorical-aware imbalance handling is required.